Room plan: shuffle machines and loop names (permutations, part 2)

The plan for Wednesday, November 4, 2026 (James confirmed shuffle machines on November 2). Week 3, Shuffle machines: packets for K–1, 2–3 and 4–5, and the facilitator guide, whose chair launch this plan uses. Unpiloted at the circle; tried at home.

Why this follows October 29

On October 29 the children counted the orders of 3 and 4 cards and built orders from swaps, with cycle notation planned but not reached. A shuffle machine is a whole order applied at once, and running it again and again splits the slots into loops. Cycle notation is just a name for those loops, so the children meet it as a way to write down something they walked through, not as a new rule.

For adults: the loops are the cycles; the number of turns until every cube is home is the least common multiple of the loop lengths (loops of 2 and 3 take 6 turns on 5 slots); a loop of k slots can be built from k − 1 swaps, which links back to last week's swaps and, for older children, to even and odd.

Prep checklist for Wednesday afternoon

Print single-sided, in colour, at 100% ("Actual size"), so the mat slots stay 1.2 inches. The guide's copy counts assume 4 children at K–1, 4 at 2–3 and 3 at 4–5; the counts below lean toward 2–3, since most of the room is there. Adjust at the door.

- K–1 packet, 10 pages: 4 copies, plus 4 extra copies of page 10 (the blank test mat) for the other tables
- Grades 2–3 packet, 11 pages: 8 copies
- Grades 4–5 packet, 8 pages: 4 copies
- Facilitator guide, 11 pages: one per adult (3)
- This room plan: one per adult (3)
- Snap cubes from school: about 75, at least 14 each of green, blue, red, yellow and purple, plus a pink one per 2–3 child and a spare (any small pink object will do); one set per child in its own cup
- Masking tape and five coloured sheets for the chairs (green, blue, red, yellow, purple)
- Coloured pencils (three colours per child at 2–3), pencils, scrap paper
- Whiteboards and markers, one per child; one big whiteboard or easel by the chairs
- Sticky notes for the floor rounds
- Two decks of playing cards for 4–5, with one suit A to Q in order on top (the guide explains why)
- A timer everyone can see
- Ten-minute check the day before or that afternoon: the guide's three tests with real cubes (cubes fit the slots; K–1 Problem 3's bottom machine takes 6 turns; the 8-card perfect shuffle)
- At the room: set up the chairs and floor arrows before the children arrive (guide, "Set up before the children arrive")

Before the children arrive

- Five chairs with their coloured sheets and masking-tape arrows on the floor, exactly as in the guide (chairs 1 → 2 → 3 → 1, chairs 4 and 5 point to themselves).
- At each table: a cup of snap cubes per child (green, blue, red, yellow, purple, pink; about 75 in all), that band's packet pages face down, coloured pencils, whiteboards.
- One timer everyone can see. A whiteboard by the chairs for loop names, and a pad of sticky notes.
- Printed early-finisher cards: "Draw a 5-slot machine that takes as many turns as you can" (2–3 and 4–5) and "Draw a machine where only one cube moves" (K–1, from its Problem 5; it can't be done).

The hour

Time	Everyone	Jobs
0:00–0:05	Arrive and run around.	
0:05–0:08	**Launch at the chairs**, as in the guide: five children sit, "switch" until everyone is home (3 switches). Practise the reset signal once.	Sitters walk; everyone else counts out loud.
0:08–0:10	**Loop names.** On the whiteboard, write the loop they just walked as (1 2 3) and say "1 goes to 2 goes to 3 goes back to 1". Chairs 4 and 5 are loops of one.	One child reads the loop name back.
0:10–0:28	**Tables, round 1**: Problem 1 in every band (run each machine until every cube is home). After each machine, the pair writes its loops on the whiteboard in loop names.	**Turner** moves the cubes; **Counter** keeps the tally and writes the loop names. Swap at every new machine.
0:28–0:33	**Back to the floor.** A pair from the 2–3 table turns one of their machines into instructions: each sitter gets a sticky note with the number of the chair they walk to (ignore the tape arrows). The room predicts the number of switches, then runs it.	New sitters; the pair is in charge of saying "switch".
0:33–0:48	**Tables, round 2**, by band. K–1: Problem 4 (draw arrows for 2, 3 and 4 turns; the partner checks with cubes). 2–3: Problem 2 (colour the loops, predict, then run), then Problem 3 (5 slots, 1 to 6 turns). 4–5: Problem 3, then "How many swaps build each of your machines?"	Same pairs, swapped jobs: the Predictor says a number before the Turner runs it.
0:48–0:55	**Floor finale: the slowest machine.** Each table brings its slowest 5-slot machine. Run the winner on the chairs with sticky notes (6 turns is the record for 5 slots).	Tables choose one sitter each.
0:55–1:00	Cubes back in cups by colour, pages collected.	Each pair has one colour to sort.

If it goes sideways

- A table finishes early: hand out its early-finisher card.
- A pair is stuck: the adult runs one turn with them and asks "Who is home now?", nothing more.
- The room gets loud: reset signal, then go straight to the next floor activity rather than talking over it.
- Jay and Rose have done the 4–5 pages at home. Make them floor captains at 0:28 and 0:48, and checkers at the 2–3 table who may say "try again" but not why.

How it went

(Add a dated line after the session.)